

Workshop 1 — End-to-End AI System Design Lab

Scenario Brief — Retail

Applies Courses 1–4. You will design a real AI system under real constraints, then defend your decisions to the trainers and to every other group in the room.

This workshop is not graded. It does not affect certification (that is Workshop 2 plus the individual quiz). It is the rehearsal. Treat it accordingly — take risks here, not in the capstone.

1. The Company

NovaMart — a large supermarket and convenience-store chain across three Asian markets. This brief concerns its merchandising and store operations.

Stores	640 (large-format supermarkets + convenience)
SKUs	42,000 chain-wide; ~11,000 per store
Transactions per day	2.9 million
Revenue	~ \$4.1 billion/year
Net margin	2.4%
Fresh / perishables	31% of revenue
Fresh waste	4.8% of fresh revenue $\approx$ \$61M/year
Out-of-stock rate	7.2% $\rightarrow$ estimated \$180M/year in lost sales
Store associates	8,400
Loyalty members	14 million (68% of transactions)
Promotions	~2,200 events/year; promo weeks distort demand 3–10×

The two numbers that decide everything

2.4% net margin. In insurance, a bad decision costs money. In grocery retail, bad *infrastructure cost* eats the profit directly. There is nowhere to hide it.

The 04:00 window. Each store's replenishment order must be finalised by **04:00** for the distribution centre to pick and the 06:00 truck to leave. Miss the window and that store **receives nothing that day**. No exceptions, no re-runs.

What the board has approved

KPI	Today	Target
Out-of-stock rate	7.2%	3.0%
Fresh waste	4.8%	3.0%
Demand-forecast accuracy (WAPE)	41%	≤ 25%
Associate time lost to lookups	~50 min/shift	< 15 min/shift
Store associate headcount	8,400	8,400 — redeployed to customer service, not reduced

The last line is deliberate. This is not a headcount programme. Designs premised on replacing associates are answering a question nobody asked.

2. The Data (identical for both variants)

POS. 5 years of history, 2.9M transactions/day — billions of rows. SKU-level, basket-level, store-level, second-level.

Loyalty. 14M members, basket history, frequency, channel.

Inventory. Stock-on-hand snapshots per store × SKU. **Read the Variant B constraints before you trust this number.**

Supplier EDI. 2,300 suppliers: lead times, case packs, price lists, delivery schedules.

Planograms. 640 stores × category, updated monthly, in PDF and CAD.

SOPs and food-safety policy. 1,400 documents, 3 languages, versioned, and **legally binding** — a wrong allergen answer is not a customer-service incident. It is a criminal matter.

Product master. 42,000 SKUs: images, attributes, shelf life, allergens, storage conditions.

Waste and shrink logs. Expiry, damage, loss.

Third-party. Weather, local events, school calendars, competitor pricing (scraped).

3. Target Capabilities

NovaMart wants these six capabilities. **How you achieve them — or whether you judge some of them not worth building — is your design decision.** Nothing here prescribes an architecture.

1. **Demand forecasting** — per SKU, per store, per day. *42,000 SKUs × 640 stores = 27 million time series.*
2. **Automated replenishment** — order quantities respecting shelf life, case packs, DC capacity, and **the 04:00 cut-off.**
3. **Markdown / dynamic pricing on fresh** — cut waste without destroying margin.
4. **Store associate copilot** — planograms, SOPs, product knowledge, food-safety policy. *"Where does this go?" "Can I still sell this?" "Does this contain peanuts?"*

- 5. **Customer service and substitutions** for online grocery orders — out of stock, so what do we swap in?
- 6. **Promotion planning support** — forecast promo lift and cannibalization.

4. Non-Functional Requirements (identical for both variants)

The batch window — the single most important constraint in this brief

- All **640 stores** × **11,000 SKUs** must be forecast *and* have orders generated **between 22:00 and 04:00**.
- A **6-hour window**. Hard wall. The window does not get longer at Lunar New Year.

Latency

- Store associate copilot: **p95 < 2 seconds** — **and it must work when the store's internet is down**. 4% of stores lose connectivity for over an hour at least weekly.
- Online order substitution: **p95 < 500ms**. The picker is standing in an aisle waiting.
- Fresh markdown decisions: computed **3× per day** (morning, midday, evening).

Availability

- **POS: 99.99%**, and it must run fully offline. POS is the revenue path. **Any AI that touches POS is a non-starter**.
- The batch path: miss 04:00 and 640 stores get no delivery.

Peak

- **Lunar New Year: 6× normal volume, for two weeks** — and the date moves in the calendar every year.
- Promo weeks distort demand 3–10×.

Cost — read this carefully

- The board approved **\$4M/year** for total AI run cost, against **\$241M** of identified upside (waste + out-of-stock).
- **\$4M/year ≈ \$10,959 per day**.
- You need **27 million forecasts per day**.
- **That is \$0.0004 per forecast**.
- Do that arithmetic *before* you choose a model. Several groups will select an architecture that is **1,000× over budget** without noticing.

Quality and safety

- Associate copilot: **food-safety and allergen questions carry criminal liability**. A hallucinated allergen answer can kill someone. Your design must state when the system **cites**, when it **refuses**, and who is accountable.
- Markdown decisions must be **auditable** — consumer-protection law prohibits arbitrary price discrimination.

Data

- One of the three markets **prohibits personal (loyalty) data leaving the country**.
- POS data is retained for 5 years.

VARIANT A — AI-First (Greenfield)

NovaMart's board has approved **NovaMart Go** — an entirely new quick-commerce format: **40 dark stores**, online-only, **30-minute delivery**, launching in Market 1. There is no legacy system. You are designing this platform from nothing, AI-First from day one.

Your constraints:

- **Scale:** 40 dark stores, **6,000 SKUs** (a much narrower assortment), targeting 25,000 orders/day by month 12.
- **Timeline:** live in **8 months**.
- **Team:** 14 engineers, 3 data scientists, 1 platform engineer. This team has to *run* whatever you design.
- **Budget:** \$3.2M capex over 18 months. AI run cost must stay **below 0.4% of GMV**.
- **The 30-minute promise is an architectural constraint, not marketing.** Bad forecast → stockout → substitution → cancelled order. That entire domino chain starts at forecast quality.
- **Cold start.** NovaMart Go has no sales history of its own. You *may* use the supermarket chain's 5 years of data — but dark-store demand behaves **completely differently**: smaller baskets, narrower assortment, a different time-of-day distribution, no browsing behaviour, and a far higher substitution rate. This is a first-class design problem, not a footnote.
- Everything in \$5 (NFRs) still applies. Same margin. Same Lunar New Year peak. Same law.

You must explicitly pass the **removal test**. Show what NovaMart Go cannot do if the AI is removed. If the answer is "it still works, just slower," you have designed an AI-Retrofit and mislabelled it.

VARIANT B — AI-Retrofit (Legacy)

You are working on the incumbent. NovaMart's merchandising and replenishment run on **MERLIN** — an SAP-based ERP, customised continuously since **2006**. On-premises. It owns merchandising, inventory, replenishment, and supplier EDI.

What MERLIN is:

- It is the **system of record** for orders sent to suppliers.
- Its replenishment engine is a **min/max reorder-point calculator from 2006** with **340,000 hand-maintained parameters**. Category managers tune them by hand. There is no automated test suite.
- MERLIN ships **twice a year**, via an SAP-certified partner. Expensively.

What surrounds it:

- **The POS estate:** 640 stores, ~7,000 lanes, a POS application from **2011** that runs **offline-first** and **syncs nightly**. **Untouchable**. Changing POS requires a 9-month re-certification with the payment processor.
-  **The truth about inventory — read this line carefully:** MERLIN's stock-on-hand is updated by the **nightly POS sync**. In-day stock levels are *estimates*. Measured perpetual-inventory accuracy is **78%**.

That means: more than one time in five, the system is wrong about what is actually on the shelf. Any design that assumes it knows real-time inventory is building on sand. You must design *for* this reality, not around it.

- **Supplier EDI:** fixed schema, 04:00 cut-off, **cannot be changed** (2,300 suppliers are integrated against it).
- **Data warehouse:** nightly ETL, up to 24 hours stale.
- **Store network:** flaky connectivity; 4% of stores lose it for over an hour weekly.
- **Replacement was attempted:** a "unified commerce platform" programme in 2021 was cancelled after two years and \$22M.

Your constraints:

- **You cannot modify MERLIN's replenishment engine.** MERLIN exposes exactly **three extension points**: (1) an **override table** MERLIN reads *before* generating orders, (2) a **review queue** for category managers *after* orders are generated, (3) the **nightly batch**. Everything you build must attach at one of these, or live entirely outside MERLIN.
- **You cannot touch POS.** Under any circumstances.
- **You cannot change the EDI schema, and you cannot move the 04:00 cut-off.**
- **Perpetual inventory accuracy is 78%.** Design for it.
- **The AI must never be able to break the ordering chain.** If your AI is unavailable, replenishment must still run — 640 stores must still get their trucks in the morning. Design for that explicitly.
- **Core replacement is off the table.** The board remembers 2021. Proposals to "gradually replace MERLIN" are proposals to be told no.
- Same **8 months**, same **\$3.2M**, same 2.4% margin.
- Everything in \$5 (NFRs) still applies.

You must produce a legacy assessment — what can change, what cannot, and what is unsafe to touch — and you must name **which of the AI-Retrofit integration patterns** you chose (sidecar, event-driven, API gateway, shadow-assist, in-process plugin) and defend why.

5. What You Submit

Six artifacts. One named owner each. Your group has 5–6 people; the submission must state who owned which artifact. This is not bureaucracy — it is how the workshop resists free-riding.

#	Artifact	What it must contain
1	C4 context + container diagram	System context and container views. Request flow narrated.
2	NFR table	Every NFR from §5, with your actual numbers — how you meet it, not that you intend to.
3	Technology-selection matrix	Weighted scoring across at least three consequential decisions. Criteria, weights, scores, and the trade-off you accepted.
4	One ADR	Your single most consequential decision. Options considered → decision → consequences.
5	Risk register	Top 5 risks. Category, likelihood × impact, mitigation, and the monitoring signal + threshold that tells you it is happening.
6	Variant-specific	Variant A: the removal test + your cold-start strategy. Variant B: the legacy assessment + your chosen integration pattern + your kill-switch design.

Format: maximum 8 slides. Not 30. If it does not fit in 8 slides, you have not made your decisions yet.

6. How You Present

Each group gets a 12-minute slot: 7 minutes to present, 4 minutes to be questioned, 1 minute to change over.

Every group in the room solved the same scenario. **Nobody re-explains the problem.** If you open with "our scenario is a supermarket chain..." you will be stopped.

7. What This Workshop Is Actually Testing

Not whether you can draw an architecture. Whether you can **defend one**.

The five things the room will be listening for:

1. Did you choose the **simplest design that meets the NFRs**, or the most impressive one?
2. Does your **cost arithmetic** survive 27 million forecasts a day? *Do the multiplication before you pick a model, not after.*
3. Did you use an **LLM for something that is not a language problem**?
4. Did you notice the **constraints that make this hard** — the 04:00 window, inventory that is only 78% accurate, stores that go offline, the allergen question, the 6× Lunar New Year peak?
5. Can you tell a **real trade-off** from one you invented to avoid doing the work?